

NEWINGTON COLLEGE



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

2003

12 MATHEMATICS

Time allowed - 3 hours

(plus 5 minutes reading time)

DIRECTIONS TO CANDIDATES:

- All questions are of equal value.
- All questions may be attempted.
- In every question, show all necessary working.
- Marks may not be awarded for careless or badly arranged work.
- Approved silent calculators may be used.
- A table of standard integrals is provided for your convenience.
- The answers to the ten questions in this paper are to be returned in separate bundles clearly marked Question 1, Question 2 etc. Start each question in a new booklet.
- Each booklet must show the candidate's computer number.
- The questions are not necessarily arranged in order of difficulty. Candidates are advised to read the whole paper carefully at the start of the examination.
- Unless otherwise stated candidates should leave their answers in simplest exact form.

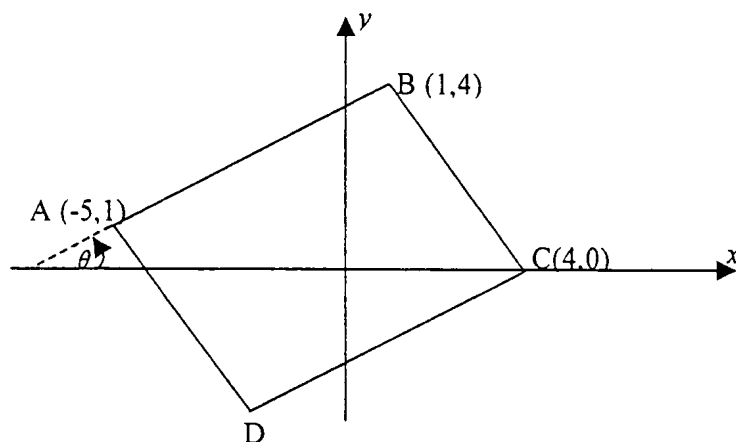
Question 1 (12 Marks)**Marks**

- Express 135° in radians in terms of π .
- Factorise $2x^2 - 18$.
- Solve for x : $|x - 3| < 4$.
- Find the exact value of $\cos \frac{7\pi}{6}$.
- Find the value of e^π , correct to 3 significant figures.
- Find integers a and b such that $(2 - \sqrt{3})^2 = a + b\sqrt{3}$.
- Simplify $\log_3\left(\frac{1}{9}\right)$.

Question 2 (12 marks)Start this question in a new booklet.

a)

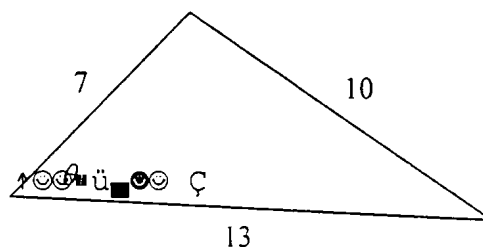
10



- Find the gradient of AB.
- If BA is extended to the x axis find θ to the nearest degree.
- Show that AB has equation $x - 2y + 7 = 0$.
- Show that the perpendicular distance from C to AB is $\frac{11}{\sqrt{5}}$ units.
- If ABCD is a parallelogram find the area of ABCD.
- If ABCD is a parallelogram find the coordinates of D.

b) Find θ to the nearest minute.

2



Question 3 (12 marks)Start this question in a new booklet.**Marks**

a) Differentiate

5

- i) $\tan 3x$
- ii) $\frac{x}{e^{2x}}$
- iii) $\log_e(\sin 3x)$

b) Find

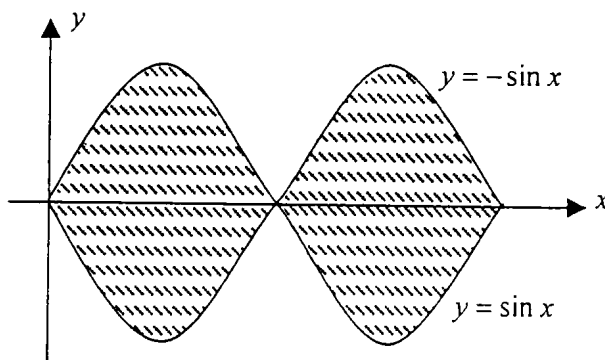
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- i) $\int \cos 3x dx$
- ii) $\int \frac{x^3 + 2x^2 - 3}{x^2} dx$

c) Find the equation of the normal to the curve $y = (3x + 1)^3$ at the point where $x = 1$.**3****Question 4 (12 marks)**Start this question in a new booklet.a) Let α and β be the roots of the equation $3x^2 + 5x - 1 = 0$.**3**

- Find
- i) $\alpha + \beta$,
 - ii) $\frac{1}{\alpha\beta}$,
 - iii) $\frac{1}{\alpha} + \frac{1}{\beta}$.

b)

**4**

Find the area bounded by $y = \sin x$ and $y = -\sin x$, as shown in the diagram above.

Question 4 continued.**Marks**

- c) A balloon accelerates vertically, so that the change of height (h) in metres is given by $\frac{dh}{dt} = 10 - \frac{1}{3}t$, where t is time in seconds after it takes off. Initially the balloon is 1 metre about the ground. 5
- i) Find the height after 6 seconds.
- ii) What is the maximum height that the balloon can reach?

Question 5 (12 marks)Start this question in a new booklet.

- a) In the Snowy Mountains the number of Corroboree frogs (Q) has been in a slow decline because of bush fires and pollution. The rate of decline is given by $\frac{dQ}{dt} = -kQ$. In 1998 the population was estimated at 2100 and in the most recent survey in 2003 it was estimated at 1050. 7
- i) Show that the function $Q = Q_0 e^{-kt}$ satisfies the equation $\frac{dQ}{dt} = -kQ$.
- ii) Find the value of the constant Q_0 .
- iii) Show that $k = 0.2 \ln 2$.
- iv) In how many years will there be only one frog remaining and hence the frogs become extinct in the Snowy Mountains?
- b) The first three terms of an arithmetic series are $25 + 19 + 13 + \dots$ 5
- i) Find the 20th term.
- ii) How many terms will it take for the sum of the terms to become negative?

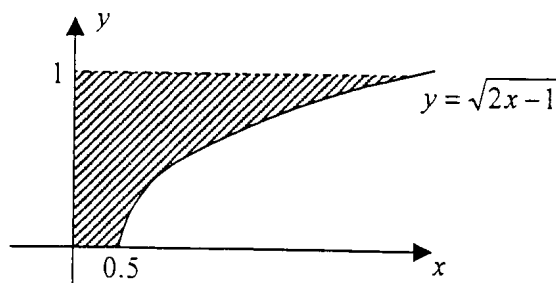
Question 6 (12 marks)Start this question in a new booklet.

Consider the curve given by $y = 1 + 3x - x^3$ for $-2 \leq x \leq 3$.

- i) Find the coordinates of the stationary points and determine their nature.
- ii) Find the point of inflexion.
- iii) Sketch the curve for $-2 \leq x \leq 3$.
- iv) What is the minimum value of y for $-2 \leq x \leq 3$?

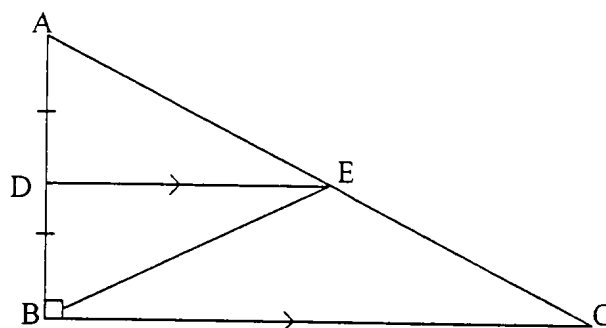
Question 7 (12 marks)Start this question in a new booklet.**Marks**

a)

4

The area shown is rotated about the y axis. Find the exact value of the volume of the solid that is generated.

b)

8

In the right angled triangle ABC , D is the mid point of AB

- i) Prove that $\angle ADE$ is a right angle.
- ii) Prove that $\triangle AED \equiv \triangle BED$.
- iii) Prove that $BE = EC$.

Question 8 (12 marks)Start this question in a new booklet.

- a)
 - i) Factorise $u^2 - 6u - 16$.
 - ii) Hence or otherwise, solve for x :

$$[\log_2 x]^2 - 6[\log_2 x] - 16 = 0.$$
- b) Find an approximate value for $\int_1^3 \log_e x dx$ using five function values and the Trapezoidal Rule. (Answers correct to 2 decimal places.)
- c) A parabola has equation: $x^2 = 8(4 - y)$.
 - Find
 - i) the focal length,
 - ii) the equation of the directrix,
 - iii) the coordinates of the focus.

Question 8 Continued

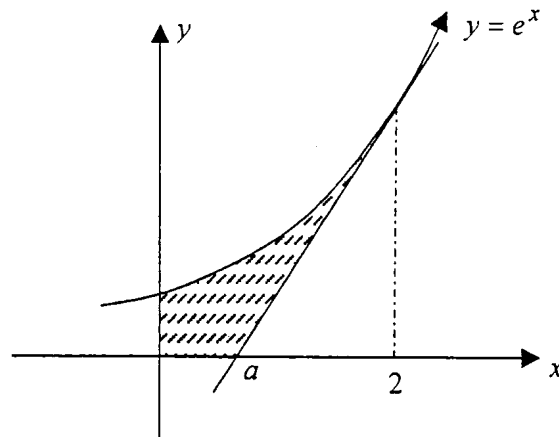
Marks

- d) i) Write down the discriminant of $3x^2 - kx + 3$. 3
ii) For what values of k does the equation $3x^2 - kx + 3 = 0$ have real and unequal roots?

Question 9 (12 marks)

Start this question in a new booklet.

- a) 6



- i) Find the equation of the tangent to $y = e^x$ at the point where $x = 2$.
ii) Show that the value of a , the point of intersect of the tangent and the x axis is 1.
iii) Find the exact size of the shaded area.
- b) Kevin and Sharon borrow \$250 000 to buy an investment property. They agree to a 15 year loan with monthly repayments and interest of 6% p.a., compounding monthly. 6
- i) Show that the amount owed after 3 months is: $A_3 = 250000(1.005)^3 - P(1.005^2 + 1.005 + 1)$, where P is the monthly repayment.
- ii) Calculate the monthly repayment to pay the loan off in the 15 years.
- iii) After 4 years, they decide to sell the property. How much is still owing on the loan?

Question 10 (12 marks)Start this question in a new booklet.**Marks**

- a) Solve the equation $2 \cos x = \sqrt{3}$, where $0 \leq x \leq 2\pi$. **2**
- b) Consider the function whose derivative is given by **3**
 $\frac{dy}{dx} = 2x^2(x-1)(2x+1)$.
Determine the nature of the stationary point at $x = 0$.
- c) You are retrenched from your position as CEO of Pear Computers. **7**
You are given a choice of “two payout” options: OPTION 1, a lump sum of \$500,000 today or OPTION 2, \$50,000 now and two equal payments of \$300,000, the first in two year’s time and the second in four year’s time.
- i) Assuming that the interest rate for the period is 6% p.a., compounding annually, show that, by the end of the 4th year, OPTION 2 is worth \$68 965.00 (nearest dollar) more than OPTION 1.
- ii) What is the minimum rate of interest that makes the OPTION 1 the better choice?

End of Paper

QUESTION ONE

(a) $\frac{3\pi}{4}$ ✓

(b) $2(x-3)(x+3)$ ✓✓

(c) The distance from x to 3 is less than 4 units i.e. $-1 < x < 7$ ✓✓

(d) $\cos \frac{7\pi}{6} = -\cos \frac{\pi}{6} = -\frac{\sqrt{3}}{2}$ ✓✓

(e) $e^\pi \approx 23.1$ ✓✓

(f) $(2-\sqrt{3})^2 = 7-4\sqrt{3}$ $a=7$ $b=-4$ ✓✓

(g) $\log_3 \frac{1}{9} = \log_3 3^{-2} = -2$ ✓

MARKING

(b) $2(x^2-9)$ ✓x

(d) One mark for knowing $\cos \frac{7\pi}{6} < 0$
One mark for knowing $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$ (e) One mark if calculation correct but
incorrectly rounded to 3 significant figures.QUESTION TWO

(a) (i) $m_{AB} = \frac{4-1}{1+5} = \frac{1}{2}$ ✓

(ii) 27° ✓

(iii) $y-4 = \frac{1}{2}(x-1)$ ✓

$2y-8 = x-1$

$x-2y+7=0$ ✓

(iv) $d = \frac{|4-0+7|}{\sqrt{1+4}} = \frac{11}{\sqrt{5}}$ ✓

(v) $d_{AB} = \sqrt{36+9} = \sqrt{45} = 3\sqrt{5}$ ✓

Area of ABCD = $\frac{11}{\sqrt{5}} \times 2\sqrt{5} = 22$ units² ✓

(vi) $M_{AC} = (-\frac{1}{2}, \frac{1}{2})$ ✓

∴ D has coordinates $(-2, -3)$ ✓✓

(b) $\cos \theta = \frac{7^2+13^2-10^2}{2 \times 7 \times 13}$ ✓

$\theta \approx 49^\circ 35'$ ✓

Question 3

a) i) $\frac{d}{dx} \tan 3x = 3 \sec^2 3x \checkmark$

1 mark answer only.

ii) $\frac{d}{dx} \frac{x}{e^{2x}} = \frac{e^{2x} \cdot 1 - x \times 2e^{2x}}{(e^{2x})^2} \checkmark$

1 mark substitution into formula.

$$= \frac{e^{2x} - 2xe^{2x}}{e^{4x}} \checkmark$$

1 mark some type of simplification.

$$= \frac{e^{2x}(1 - 2x)}{e^{4x}}$$

= 2 marks.

$$= \frac{1 - 2x}{e^{2x}}$$

iii) $\frac{d}{dx} \log_e (\sin 3x) = \frac{1}{\sin 3x} \times 3 \cos 3x \checkmark$
 $= \frac{3 \cos 3x}{\sin 3x} \checkmark$

1 mark the fraction

* 1 mark was deducted from a student who added "C" to all answers.

1 mark the correct differentiation
(2 marks).

b) i) $\int \cos 3x \, dx = \frac{1}{3} \sin 3x + C \checkmark$

1 mark integral
C was not needed for the mark

ii) $\int \frac{x^3 + 2x^2 - 3}{x^2} \, dx$

$$= \int \frac{x^3}{x^2} + \frac{2x^2}{x^2} - \frac{3}{x^2} \, dx \checkmark$$

1 mark for separating into correct fractions or using factors.

$$= \int x + 2 - \frac{3}{x^2} \, dx$$

Question 3

$$= \frac{x^2}{2} + 2x + \frac{3}{x} + C$$

1 mark integrations.
all correctly done.

1 mark for $+C$
with 2 correct
integrations.

= 3 marks

$$(c) y = (3x+1)^3$$

$$\frac{dy}{dx} = 3(3x+1)^2 \times 3$$

$$\therefore \frac{dy}{dx} = 9(3x+1)^2 \quad \checkmark$$

1 mark for the differentiation

gradient of tangent at $x=1$

$$\begin{aligned} m &= 9(3 \times 1 + 1)^2 \\ &= 9 \times 16 \\ &= 144 \end{aligned}$$

gradient of normal at $x=1$

$$m = \frac{-1}{144}$$

1 mark for the gradient
of the normal

equation of the normal

$$y - y_1 = m(x - x_1)$$

$$y = (3 \times 1 + 1)^3$$

$$y = 64$$

$$y - 64 = \frac{-1}{144}(x - 1) \quad \checkmark$$

1 mark correct
substitution into the
formula

= (3 marks)

$$144y - 9216 = -x + 1$$

$x + 144y - 9217 = 0$ is the
required equation.

Question 4

a) $3x^2 + 5x - 1 = 0$

$$(i) \alpha + \beta = -\frac{b}{a} \\ = -\frac{5}{3} \quad (1)$$

1 mark each

$$(ii) \alpha\beta = \frac{c}{a} \\ = -\frac{1}{3}$$

$$\frac{1}{\alpha\beta} = -3 \quad (1)$$

$$(iii) \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} \\ = \frac{-\frac{5}{3}}{-\frac{1}{3}} \quad (1) \\ = 5$$

(b) $A = 4 \int_0^\pi \sin x \, dx \quad (1) \quad (1)$

$$= 4 \left[-\cos x \right]_0^\pi \quad (1)$$

$$= 4 \left[-\cos \pi + \cos 0 \right]$$

$$= 4 \left[+1 + 1 \right] \quad (1)$$

$$= 8 \text{ sq units}$$

1 mark correct integrand

1 mark multiplying effect + limits

1 mark integration

1 mark evaluation.

(c) $\frac{dh}{dt} = 10 - \frac{1}{3}t$

$$h = 10t - \frac{1}{3} \frac{t^2}{2} + c \quad (1)$$

$$h = 10t - \frac{t^2}{6} + c$$

when $t = 0$ $h = 1$

$$1 = c$$

Question 4

$$\therefore h = 10t - \frac{t^2}{6} + 1$$

$$h = ? \quad t = 6$$

$$h = 10 \times 6 - \frac{6^2}{6} + 1$$

$$= 60 - 6 + 1$$

$$\therefore \underline{h = 55 \text{ m}} \quad (1)$$

$$11) \quad h = 10t - \frac{t^2}{6} + 1$$

$$\frac{dh}{dt} = 10 - \frac{1}{3}t$$

$$10 - \frac{1}{3}t = 0 \quad (1)$$

1 mark showing $\frac{dh}{dt} = 0$

$$30 - t = 0$$

$$t = 30$$

$$\frac{d^2h}{dt^2} = -\frac{1}{3}$$

$$< 0$$

(1)

1 mark showing it is a max

$$\therefore \text{max at } t = 30$$

$\therefore$ max height

$$h = 10 \times 30 - \frac{30^2}{6} + 1$$

$$= 300 - 150 + 1$$

$$= 151 \text{ m}$$

(1)

1 mark for answer.

$$\therefore \text{max height is } 151 \text{ m.}$$

Question 5

a) i) $Q = Q_0 e^{-kt}$

$$\begin{aligned}\frac{dQ}{dt} &= \frac{d}{dt}(Q_0 e^{-kt}) \\ &= Q_0 \times -k e^{-kt} \quad \checkmark \\ &= -k Q_0 e^{-kt} \\ &= -k Q \quad \checkmark \quad (2)\end{aligned}$$

ii) $t=0 \quad Q=2100$

$$\begin{aligned}2100 &= Q_0 e^{-k \times 0} \quad \checkmark \\ \therefore Q_0 &= 2100 \quad (1)\end{aligned}$$

iii) $t=6 \quad Q=1050$

$$\begin{aligned}1050 &= 2100 e^{-6k} \\ \frac{1}{2} &= e^{-6k} \quad \checkmark \\ -6k &= \ln \frac{1}{2} \\ 6k &= \ln 2 \\ k &= \frac{1}{6} \ln 2 \quad \checkmark \quad (2) \\ &= 0.2 \ln 2\end{aligned}$$

must show

$$\ln \frac{1}{2} = -\ln 2$$

iv) $1 = 2100 e^{-0.2 \ln 2 t}$

$$e^{-0.2 \ln 2 t} = \frac{1}{2100} \quad \checkmark$$

$$-0.2 \ln 2 t = -\ln 2100$$

$$\begin{aligned}t &= \frac{\ln 2100}{0.2 \ln 2} \quad \checkmark \\ &\approx 55 \text{ years} \quad (2)\end{aligned}$$

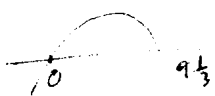
b) i) $a=25, d=-6 \quad \checkmark$

$$\begin{aligned}T_{20} &= 25 + 19 \times -6 \quad \checkmark \quad (2) \\ &= -89\end{aligned}$$

ii) $S_n < 0 \quad \frac{n}{2}(50 + (n-1) \times -6) < 0 \quad \checkmark$

$$\begin{aligned}n(56 - 6n) &< 0 \\ n < 0 \text{ or } n > 9\frac{1}{3} \quad \checkmark\end{aligned}$$

$$\therefore n = 10 \quad (n \neq 0) \quad \checkmark \quad (3)$$



Q6

i) $y' = 3 - 3x^2$ ✓

stationary points $y' = 0$

$$0 = 3(1 - x^2) \quad ✓$$

$$x = \pm 1$$

$$x = 1 \quad y = 1 + 3 - 1 = 3$$

$$x = -1 \quad y = 1 - 3 + 1 = -1$$

$$(1, 3) \quad (-1, -1) \quad ✓✓$$

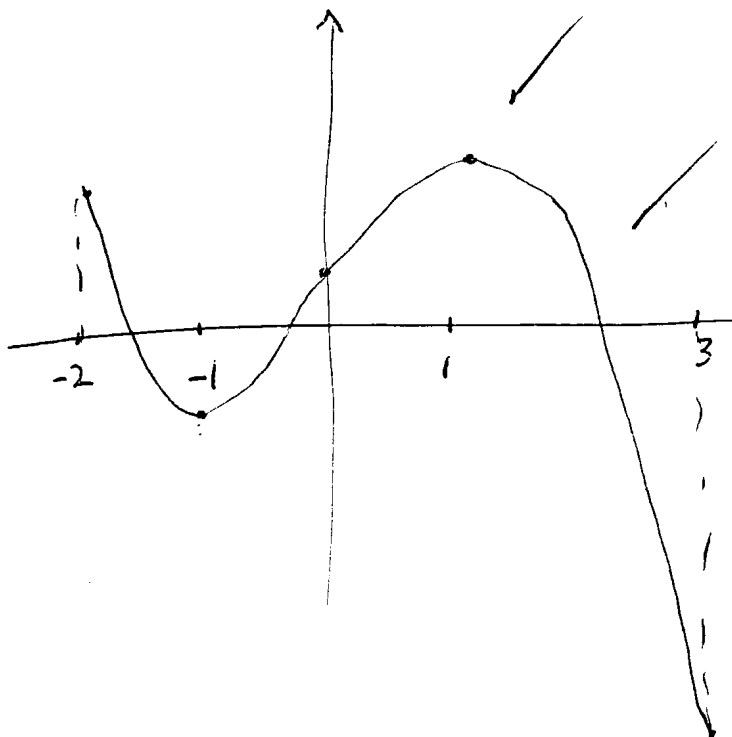
$$y'' = -6x$$

$$x = 1 \quad y'' = -6 < 0$$

$$\therefore \text{max } (1, 3) \quad ✓$$

$$x = -1 \quad y'' = 6 > 0$$

$$\therefore \text{min } (-1, -1) \quad ✓$$



iv) 17 ✓

ii) $0 = -6x$
 $x = 0 \quad y = 1$
 $(0, 1) \quad ✓$

check

x	$-\frac{1}{2}$	0	$\frac{1}{2}$
y''	3	0	-3

 ✓

change in concavity.

$\therefore$ point of inflexion

iii) $x = -2 \quad y = 1 - 6 + 8 = 3$
 $(-2, 3) \quad ✓$

$$x = 3 \quad y = 1 + 9 - 27 = -17$$

$$(3, -17)$$

$$7/a) \quad y = \sqrt{2x} - 1$$

$$y^2 + 1 = 2x$$

$$x = \frac{(y^2 + 1)^2}{4}$$

$$V = \pi \int_0^1 x^2 dy$$

$$= \frac{\pi}{4} \int_0^1 (y^4 + 2y^2 + 1) dy$$

$$= \frac{\pi}{4} \left[\frac{1}{5} y^5 + \frac{2}{3} y^3 + y \right]_0^1$$

$$= \frac{\pi}{4} \left(\frac{1}{5} + \frac{2}{3} + 1 \right)$$

$$= \frac{7\pi}{15} \text{ units}^3$$

(b) (i) $\hat{ADE} = \hat{ABC}$ (corresponding $\angle$ s, $DE \parallel BC$)
 $= 90^\circ$

(ii) $\hat{BDE} = 180^\circ - \hat{ADE}$ (ADB straight $\angle$)
 $= 90^\circ$

DE is common

$AD = BD$ (given)

$\therefore \triangle AED \equiv \triangle BED$ (SAS)

(iii) $\hat{AED} = \hat{ECB}$ (corresponding $\angle$ s, $DE \parallel BC$)

$\hat{BED} = \hat{ECB}$ (alternate $\angle$ s, $DE \parallel BC$)

but $\hat{AED} = \hat{BED}$ (equal corresponding $\angle$ s in congruent $\triangle$ s AED , BED)

$\therefore \hat{ECB} = \hat{ECB}$

$\therefore \triangle EBC$ is isosceles (equal base $\angle$ s)

$\therefore BE = EC$

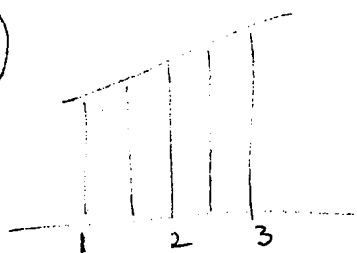
8/ (u) (i) $(u-8)(u+2)$
 (ii) ~~Let $u = \log_2 x$~~

$$\log_2 x = 8 \quad \text{or} \quad \log_2 x = -2$$

$$x = 2^8 \\ = 256$$

$$x = 2^{-2} \\ = \frac{1}{4}$$

(b)

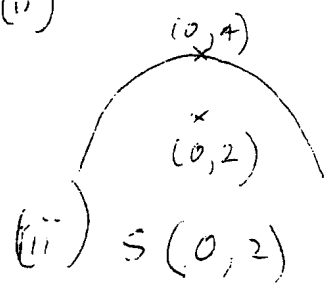


$$\begin{aligned} \int_1^3 \log x \, dx &= \frac{1}{4} [f(1) + f(3) + 2[f(1.5) + f(2) + f(2.5)]] \\ &= \frac{1}{4} (0 + \log 3 + 2(\log 1.5 + \log 2 + \log 2.5)) \\ &= \frac{1}{4} \log 168.75 \\ &= 1.28 \text{ (2 decimal places)} \end{aligned}$$

(c)

(i) 2

(ii) $y = 6$



(d) (i) $k^2 - 4.3.3$

(ii) $k^2 - 36 > 0$

$k < -6$ or $k > 6$

QUESTION 9

a) i) $\frac{dy}{dx} = e^x$

- $x = e^2$ at $(2, e^2)$

$y - e^2 = e^2(x - 2)$

$y - e^2 = e^2x - 2e^2$

$e^2x - y - e^2 = 0$

(or $y = e^2x - e^2$)

ii) when $y = 0$

$e^2x - e^2 = 0$

$\therefore x = 1$

iii) Shaded Area = $\int_0^2 e^x dx - \frac{1}{2} \times 1 \times e^2$
(or $\int_0^1 e^x dx - \int_1^2 (e^2x - e^2) dx$)

$= [e^x]_0^2 - \frac{1}{2}e^2$

$= e^2 - 1 - \frac{1}{2}e^2$

$= \frac{1}{2}e^2 - 1 \text{ units}^2$

b) i) A_n = amount owed after n months $r = \frac{0.06}{12} = 0.005$

$A_1 = 250000 \times 1.005 - P$

$A_2 = (250000 \times 1.005 - P) \times 1.005 - P$

$= 250000 \times 1.005^2 - P(1 + 1.005)$

$\therefore A_3 = 250000 \times 1.005^3 - P(1 + 1.005 + 1.005^2)$

ii) $n = 15 \times 12 = 180 \text{ months}$ $A_{180} = 0$

$A_{180} = 250000(1.005)^{180} - P(1 + 1.005 + \dots + 1.005^{179})$

$250000(1.005)^{180} = P \left(\frac{1(1.005)^{180} - 1}{1.005 - 1} \right)$

$P = 250000(1.005)^{180} \times \frac{0.005}{1.005^{180} - 1}$

$= \underline{\underline{\$2109.64}}$

iii) 4 yrs = 48 mth

$A_{48} = 250000(1.005)^{48} - 2109.64 \left(\frac{1.005^{48} - 1}{0.005} \right)$

$= \$203475.23$

1 - find gradient

1 - equation of line

1 - attempt to solve for x when $y = 0$.

2 - correct area expression (2 parts)

NOTE: 1 mark given for $\int_0^2 e^x - (e^2x - e^2) dx$

1 - answer in exact form

1 - adjust rate

1 - derive expression showing expansion.

1 - $A_{180} = 0$

1 - GP sum

1 - correct solution.

1 - correct subst.

QUESTION 10

a) $2\cos x = \sqrt{3}$

$\cos x = \frac{\sqrt{3}}{2}$

related $x = \pi/6$

* $\therefore x = \pi/6, 11\pi/6$

b) $\frac{dy}{dx} = 2x^2(x-1)(2x+1) (= 2x^2(2x^2-x-1))$

$\frac{d^2y}{dx^2} = 4x(2x^2-x-1) + 2x^2(4x-1)$

when $x=0$, $\frac{d^2y}{dx^2} = 0$

 $\therefore$ possible horizontal inflexion.

stat pts $\Rightarrow 0, 1, -1/2$

check

x	$-1/4$	0	$1/2$
$\frac{d^2y}{dx^2}$	$-$	0	$-$

 $\therefore$ horizontal inflexion at $x=0$.

c) i) Option 1 $\Rightarrow A_4 = 500\,000 \times 1.06^4$
 $= \$631\,238.48$

Option 2

$$A_4 = 50\,000 \times 1.06^4 + 300\,000 \times 1.06^2 + 300\,000$$
$$= \$700\,203.85$$

Opt 2 - Opt 1 = $\$68\,965$

ii) $500\,000r^4 > 50\,000r^4 + 300\,000r^2 + 300\,000$

$450\,000r^4 - 300\,000r^2 - 300\,000 > 0$

$45r^4 - 30r^2 - 30 > 0$

let $u = r^2$

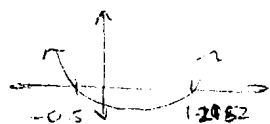
$$u = \frac{30 \pm \sqrt{900 + 5400}}{2 \times 45}$$

$$= \frac{30 \pm 30\sqrt{7}}{90}$$

$$= \frac{1 \pm \sqrt{7}}{3}$$

$\approx 1.2152 \text{ or } -0.5155 \dots$ (reject negative)

$\therefore r = \sqrt{1.2152}$
$$= 1.1023$$

 $\therefore$ minimum rate is 10.2%

1 - related angle

1 - correct quadrants & expressed in radians.

1 - $\frac{d^2y}{dx^2} = 0$ or check any points1 - check pts between stationary points

1 - correct conclusion from info gained.

1 - calculate Opt 1

1 - expression for Opt 2

1 - calculate Opt 2

1 - subtraction.

1 - inequality statement

1 - solve quadratic

1 - correct answer